# Updated Lamport-style Proof Supplement

Metropolis Random Walks Escape Deep Basins
Faster in Higher Dimensions

Companion to the latest supplied LaTeX manuscript

Approved manuscript and synchronized supplement, 10 November 2026

**Purpose.** This supplement renders the arguments of the corrected manuscript as hierarchical claims with explicit justifications and dependencies. It accompanies the 10 November 2026 source revision and supersedes the earlier supplements for that source.

**Revision summary.** This edition incorporates the approved wording, attribution and layout corrections. Escape is a first hit of any site on the outer shell. The introduction's lower bound is explicitly a resistance-based order bound. Section V-C uses Dirichlet's potential principle; V-E retains Thomson's flow principle. Remark 2 limits the unbiased identity to the quotient. Related work is narrowed to supported claims without treating the general Bovier framework as restricted to fixed landscapes. Figures 1–2 and numerical float placement are improved; long displays are split. The mathematical proof route is unchanged.

**Verification status.** The former GAP-001 is closed by the geometric-tail argument explicitly added to the source. The exact external countable-state and infinite-network background remains conditional under this review's source boundary, as do the ancillary external comparisons. No counterexample or invalid central counting, transport, conservation, or energy calculation was identified.

**Method and confidence.** The conversion, forward audit, and reverse audit follow the WWresearch/lamport-proof v0.2.0 workflow. These are instructions for structured mathematical review, not a theorem checker. One assistant performed the passes sequentially; no independent reviewer or proof-assistant kernel checked the result. The companion exact-rational script passes 176 finite diagnostic cases.

**Coverage.** All four theorems, both propositions, both lemmas, shell counting and proposal averages, low-dimensional substitutions, the unbiased quotient identity, Remark 3, and the symmetry basis of the numerical method. The new $p = 1, N = 40$ examples are independently reproduced by `check_unbiased.py`; they are finite numerical diagnostics, not proof premises. Other numerical datasets, empirical claims and literature summaries are outside this proof audit. Commented-out source is excluded.

# Contents


**I Source-mapped structured proofs** **3**

**1 Rendering status** **3**

**2 Frozen source contract** **3**

**3 Source inventory** **3**

**4 Lamport-style rendering** **4**
4.1 Geometry and shell probabilities . . . 4
4.2 Theorem 1: exact quotient escape time . . . 5
4.3 Special cases and Remark 2 . . . 6
4.4 Proposition 1: reversible shell quotient . . . 6
4.5 Theorem 2: quotient asymptotic order . . . 7
4.6 Remark 3: sharper leading constant . . . 8
4.7 Lemma 1: hitting time via capacity . . . 8
4.8 Theorem 3: true-chain lower bound . . . 10
4.9 Lemma 2: transport on nested cubes . . . 11
4.10 Proposition 2: resistance upper bound . . . 12
4.11 Theorem 4: true-chain asymptotic scaling . . . 13
4.12 Ancillary symmetry and numerical-method identities . . . 14

**5 Source-to-step mapping ledger** **15**

**6 Gap and external-support register** **17**

**7 Frozen audit handoff** **18**

**II Forward audit** **18**

**8 Verdict and theorem contracts** **18**

**9 Proof structure and findings** **18**

**10 Step ledger** **19**

**11 Exact-arithmetic diagnostics** **20**

**12 Minimal clarification plan** **20**

**III Reverse dependency audit** **21**

**13 Audit boundary** **21**

**14 Obligation graph** **21**

**15 Critical findings** **21**

**16 Dependency table** **21**

**17 Verdict** **23**

**A Repository use and provenance** **23**

# Part I

# Source-mapped structured proofs

## 1 Rendering status

**SOURCE-MAPPED** for the declared mathematical scope. This status concerns source mapping only; it is not a validity verdict. Local proof steps are normalized with the new leading-constant proof explicitly mapped to the corrected source. Unavailable external theorem statements remain marked.

## 2 Frozen source contract

The authoritative input is the corrected `source/paper.tex` included with the repository package, derived from the supplied `paper (2).tex` with the authorized revisions. Its SHA-256 is

`1e6e499cb06f1df7b60044440f194358`
`0feba5bbfcc039180bd8e17f86ad7182`

$D, N$ are positive integers. Exact quotient formulas allow $0 < p \leq 1$, with $\alpha = 1/p$; deep-basin and resistance results assume fixed $D$ and fixed $0 < p < 1$. States lie in $\mathbb{Z}^D$, energy is $V(x) = \|x\|_\infty$, and proposals are uniform over $2D$ nearest neighbors. An uphill proposal is accepted with probability $p$; other moves are accepted. Time counts attempted moves, including rejections. Escape is first arrival anywhere on $A_N = \{x : \|x\|_\infty = N\}$.

$Q_D(N)$ denotes the mean for the separately defined shell-quotient chain; $H_D(N) = \mathbb{E}_0\tau_N$ denotes the true mean. They agree in dimension one; no general inequality $Q_D(N) \leq H_D(N)$ is assumed or proved here. Quotient resistance is a lower bound on true resistance, which is a different assertion.

Use the unnormalized measure $\mu(x) = p^{\|x\|_\infty}$ and conductance $c(x, y) = \mu(x)P(x, y)$. Dirichlet energy is $\frac{1}{2}\sum_{x\sim y} c(x, y)(f(x) - f(y))^2$, counting both orientations. Flow energy counts each unoriented edge once; divergence is outgoing signed current. Self-loops affect holding times but have zero energy. Capacity is the infimum of energy over $f(a) = 1$, $f|_A = 0$. The transport lemma's $u_k$ denotes a distribution; the shell calculation's $u_k$ denotes a scalar proposal probability.

**Background boundary.** Finite counting, algebra, elementary positive-series/ratio arguments and explicitly stated finite-chain first-step identities are checked locally. The exact texts behind the invoked countable-state recurrence theorem, Dirichlet capacity/resistance identity and Thomson flow inequality for infinite networks with finite targets, Kemeny–Snell aggregation criterion, and predecessor parameter comparison were not supplied or newly verified. They remain EXT-001 through EXT-004, rather than assumed verified because they are familiar.

## 3 Source inventory

Source locations use stable section names and LaTeX labels. Numbers (2)–(31) follow the paper's existing equation sequence; labels are authoritative if recompilation changes numbering. Each unit contract and each numbered step has a corresponding entry in the machine-readable ledger. The companion `source_map.json` also records exact source-label anchors and their one-based line numbers for every unit and step.

| Unit | Source segment and role | Placement / source label |
|---|---|---|
| G | III-A, III-C; Eqs. (2)–(9); Geometry and shell probabilities | eq:MDs, eq:SD, eq:uk, eq:vk |
| T1 | IV-A–B; Eqs. (10)–(17); Theorem 1: exact quotient escape time | thm:exact-escape, eq:T0-exact |
| S | IV-C; Eqs. (18)–(20), Remark 2; Special cases and Remark 2 | eq:T0-1D, eq:T0-2D, eq:T0-3D, rmk:unbiased |

| Unit | Source segment and role | Placement / source label |
|---|---|---|
| P1 | V-A; Eqs. (21)–(22); Proposition 1: reversible shell quotient | prop:quotient, eq:conductance |
| T2 | V-B; Eqs. (23)–(24); Theorem 2: quotient asymptotic order | thm:asymptotic, eq:theta-scaling |
| R3 | V-B; Eq. (25); Remark 3: sharper leading constant | rmk:sharp, eq:sharp-asymptotic |
| L1 | V-C; Eqs. (26)–(27); Lemma 1: hitting time via capacity | lem:hitting, eq:hitting-identity, eq:hitting-bounds |
| T3 | V-D; Eq. (28); Theorem 3: true-chain lower bound | thm:lower-bound, eq:lower-bound |
| L2 | V-E; Eq. (29); Lemma 2: transport on nested cubes | lem:transport, eq:transport-energy |
| P2 | V-E; Eq. (30); Proposition 2: resistance upper bound | prop:resistance-upper, eq:Reff-upper |
| T4 | V-E; Eq. (31); Theorem 4: true-chain asymptotic scaling | thm:true-scaling, eq:true-scaling |
| N | II; III-C; IV-C Remark 1; VI-A; Ancillary symmetry and numerical-method identities | II aggregation; III-C; rmk:1D; VI-A |

# 4 Lamport-style rendering

## 4.1 Geometry and shell probabilities

**ASSUME / PROVE.** For integers $D \geq 1$, $k \geq 1$, the shell cardinality and the uniform shell-averaged proposal probabilities are $S_D(k) = (2k+1)^D - (2k-1)^D$, $u_k = (2k+1)^{D-1}/S_D(k)$, and $v_k = (2k-1)^{D-1}/S_D(k)$. Set $S_D(0) = 1$ separately.

**Source:** III-A, III-C; Eqs. (2)–(9). **Forward:** PASS. **Reverse:** FOLLOWS.

**G.1.** The number of points with exactly $s$ boundary coordinates is $M_{D,s}(k) = \binom{D}{s} 2^s (2k-1)^{D-s}$, $1 \leq s \leq D$.

*Justification.* Choose the $s$ coordinates, their signs, and then the interior values of the remaining coordinates. These choices identify each point exactly once.

Source: III-A; (2); SRC-001.

**G.2.** $S_D(k) = \sum_{s=1}^{D} M_{D,s}(k) = (2k+1)^D - (2k-1)^D$.

*Justification.* The classes in G.1 partition the shell. Equivalently, subtract the inner cube from the outer cube. The source assigns the origin its separate cardinality 1.

Source: III-A; (3); SRC-002.

**G.3.** The outward average is $u_k = (2k+1)^{D-1}/S_D(k)$.

*Justification.* Each point in class $s$ has $s$ outward directions among $2D$. Apply the following source-supplied binomial calculation to the average.

Source: III-C; (6)–(8); SRC-003.

**G.3.1.** $\sum_{s=1}^{D} s\binom{D}{s} 2^s a^{D-s} = 2D(a+2)^{D-1}$.

*Justification.* Differentiate the binomial expansion of $(a+2)^D$, multiply the derivative by $a$, and subtract it from $D(a+2)^D$. This is a polynomial identity, including $D = 1$.

Source: III-C; (7); SRC-004.

**G.3.2.** With $a = 2k - 1$, division by $2DS_D(k)$ gives the claim G.3.

*Justification.* Substitute G.3.1 into the average in (6); $D \geq 1$ and $S_D(k) > 0$ justify division.

Source: III-C; (8); SRC-005.

**G.4.** The inward average is $v_k = (2k-1)^{D-1}/S_D(k)$.

*Justification.* Only $s = 1$ permits a move to shell $k - 1$. Such a point has one inward direction. Use $M_{D,1}(k) = 2D(2k-1)^{D-1}$ and divide by $2DS_D(k)$.

Source: III-C; (9); SRC-006.

**G.5.** The quotient probabilities are $pu_k$, $v_k$, and $r_k = 1 - pu_k - v_k$.

*Justification.* Outward proposals are accepted with probability $p$; inward proposals are accepted with probability one. All remaining attempted moves leave the shell unchanged. These probabilities are an average over a finite shell, so the residual is nonnegative.

Source: III-C; after (9); SRC-007.

**G.6.** Q.E.D. for the stated shell cardinalities and proposal averages.

*Justification.* By G.2–G.5. These are averaged coefficients, not an assertion of Markov lumpability.

Source: III-C; end of calculation; SRC-008.

## 4.2 Theorem 1: exact quotient escape time

**ASSUME / PROVE.** Fix integers $D, N \geq 1$, $0 < p \leq 1$, and $\alpha = 1/p$. Let $T_k$ be the quotient-chain mean hitting time of $N$, and $Q_D(N) = T_0$. Then

$$Q_D(N) = \sum_{\ell=0}^{N-1} \frac{\alpha^{\ell+1} + \sum_{m=1}^{\ell} S_D(m)\alpha^{\ell-m+1}}{(2\ell+1)^{D-1}}.$$

**Source:** IV-A–B; Eqs. (10)–(17). **Forward:** PASS WITH MINOR ISSUES. **Reverse:** FOLLOWS.

**T1.1.** Define $b_k = T_k - T_{k+1}$ for $0 \leq k < N$. Then $b_0 = 1/p$ and $pu_k b_k = v_k b_{k-1} + 1$ for $1 \leq k < N$, with $T_N = 0$.

*Justification.* Use the source first-step equations $T_k = 1 + pu_k T_{k+1} + v_k T_{k-1} + r_k T_k$ and $T_0 = 1 + pT_1 + (1-p)T_0$, and rearrange. Finite-mean absorption is an implicit finite-chain prerequisite; see the audit note A-01.

Source: IV-A; (10)–(12); SRC-009.

**T1.2.** For $R_k = v_k/(pu_k) = \alpha((2k-1)/(2k+1))^{D-1}$,

$$b_k = \left(\prod_{j=1}^{k} R_j\right) b_0 + \sum_{m=1}^{k} \frac{1}{pu_m} \prod_{j=m+1}^{k} R_j.$$

*Justification.* The source unrolls the recurrence for $k = 1, 2, 3$ and invokes induction for the displayed formula. The induction is on the transient index $k$; empty products are one and empty sums zero.

Source: IV-A; (13)–(14); SRC-010.

**T1.3.** For $0 \leq m < k$, $\prod_{j=m+1}^{k} R_j = \alpha^{k-m}((2m+1)/(2k+1))^{D-1}$.

*Justification.* The factors $(2j-1)/(2j+1)$ cancel successively. The source states the formula for $k \geq m+1$; its use at $k = m$ in the sum is the empty-product convention, noted in A-01.

Source: IV-B; (15); SRC-011.

**T1.4.** The initial contribution is $\alpha^{k+1}/(2k+1)^{D-1}$; the contribution indexed by $m \geq 1$ is $S_D(m)\alpha^{k-m+1}/(2k+1)^{D-1}$.

*Justification.* Insert $b_0 = 1/p$ and $1/(pu_m) = S_D(m)/(p(2m+1)^{D-1})$ into T1.2. Use T1.3 to cancel the positive factors.

Source: IV-B; terms preceding (16); SRC-012.

**T1.5.** $b_k = (2k+1)^{-(D-1)}[\alpha^{k+1} + \sum_{m=1}^{k} S_D(m)\alpha^{k-m+1}]$.

*Justification.* Add the two contributions in T1.4. This also covers $k = 0$.

Source: IV-B; (16); SRC-013.

**T1.6.** Q.E.D.: $Q_D(N) = \sum_{k=0}^{N-1} b_k$ has the asserted formula.

*Justification.* Telescope $T_k - T_{k+1}$ and use $T_N = 0$. For $N = 1$ the sum is $b_0 = 1/p$; no interior recurrence is needed.

Source: IV-B; (17); SRC-014.

## 4.3 Special cases and Remark 2

**ASSUME / PROVE.** The quotient formula specializes to $S_1(m) = 2$, $S_2(m) = 8m$, $S_3(m) = 24m^2+2$ for $m \geq 1$. For $p = 1$, $Q_D(N) = N^2$ for all $D$; this identity is not asserted for the true-chain mean $H_D(N)$.

**Source:** IV-C; Eqs. (18)–(20), Remark 2. **Forward:** PASS. **Reverse:** FOLLOWS.

**S.1.** The three special-case formulas are obtained by replacing $S_D(m)$ by 2, $8m$, or $24m^2+2$ and the denominator by 1, $2\ell+1$, or $(2\ell+1)^2$.

*Justification.* Expand the powers in G and substitute into Theorem 1. This is an algebraic specialization, not an independent numerical validation.

Source: IV-C; (18)–(20); SRC-015.

**S.2.** If $p = 1$, then $1 + \sum_{m=1}^{\ell} S_D(m) = (2\ell+1)^D$ and $b_\ell = 2\ell+1$.

*Justification.* The successive odd powers cancel in the shell sum. Substitute $\alpha = 1$ into T1.

Source: IV-C; Remark 2; SRC-016.

**S.3.** Q.E.D.: $Q_D(N) = \sum_{\ell=0}^{N-1}(2\ell+1) = N^2$.

*Justification.* Sum the first $N$ odd integers. The revised Remark 2 explicitly restricts this dimension-independent equality to the quotient chain and warns that $H_D(N)$ need not equal $N^2$.

Source: IV-C; Remark 2; SRC-017.

## 4.4 Proposition 1: reversible shell quotient

**ASSUME / PROVE.** On $B_N = \{x : \|x\|_\infty \leq N\}$, reject proposals leaving the cube. The resulting un-killed chain is reversible with measure $\mu(x) = p^{\|x\|_\infty}$. Shorting shells while retaining shell masses gives transitions $pu_k, v_k$ for $1 \leq k < N$ and outward probability $p$ at 0. Stop either chain only upon hitting $A_N$.

**Source:** V-A; Eqs. (21)–(22). **Forward:** PASS. **Reverse:** FOLLOWS.

**P1.1.** $\mu(x)P(x,y) = \mu(y)P(y,x)$ for all pairs.

*Justification.* If $V(y) = V(x)+1$, both sides equal $p^{V(y)}/(2D)$. For equal-energy neighbors the proposal probabilities and weights agree. The reverse case follows by symmetry; nonedges have zero off-diagonal transition probabilities and self-loops balance themselves.

Source: V-A; detailed balance; SRC-018.

**P1.2.** The mass of $A_k$ is $S_D(k)p^k$, and the cut conductance is $C_k = p^{k+1}(2k+1)^{D-1}$.

*Justification.* All vertices on $A_k$ have weight $p^k$. There are $2D(2k+1)^{D-1}$ oriented outward edges, each with conductance $p^{k+1}/(2D)$. The origin is handled in P1.4.

Source: V-A; shell mass and edge count; SRC-019.

**P1.3.** For $1 \le k < N$, $\widetilde{P}(k,k+1) = C_k/(S_D(k)p^k) = pu_k$ and $\widetilde{P}(k,k-1) = C_{k-1}/(S_D(k)p^k) = v_k$.

*Justification.* Divide the retained intershell conductance by the retained shell mass, using $p > 0$ and $S_D(k) > 0$.

Source: V-A; after (22); SRC-020.

**P1.4.** $C_0 = p$ and $\widetilde{P}(0,1) = p$; internal moves become a self-loop.

*Justification.* The $2D$ edges from the origin have conductance $p/(2D)$ and the origin has mass one. Retaining all shell mass retains rejected proposals and internal transitions in the diagonal.

Source: V-A; final proof paragraph; SRC-021.

**P1.5.** Q.E.D. for reversibility and the stopped quotient construction.

*Justification.* Use P1.1–P1.4. Reflection outside $B_N$ changes no transition before reaching its boundary. Absorption itself is not claimed reversible.

Source: V-A; proposition statement and close; SRC-022.

## 4.5 Theorem 2: quotient asymptotic order

> **ASSUME / PROVE.** For fixed integer $D \ge 1$ and fixed $0 < p < 1$, with $\alpha = 1/p > 1$, $Q_D(N) = \Theta(\alpha^N/N^{D-1})$ as $N \to \infty$.

**Source:** V-B; Eqs. (23)–(24). **Forward:** PASS WITH MINOR ISSUES. **Reverse:** FOLLOWS.

**T2.1.** Define $F_\ell = 1 + \sum_{m=1}^{\ell} S_D(m)\alpha^{-m}$. Then $b_\ell = \alpha^{\ell+1}F_\ell/(2\ell+1)^{D-1}$.

*Justification.* Factor $\alpha^{\ell+1}$ from the exact expression.

Source: V-B; (24); SRC-023.

**T2.2.** $1 \le F_\ell \uparrow F_\infty < \infty$, and $b_\ell \asymp \alpha^\ell/\ell^{D-1}$ as $\ell \to \infty$.

*Justification.* The shell polynomial is of order $m^{D-1}$, and its product with $\alpha^{-m}$ is summable because $\alpha > 1$. All summands are nonnegative. The comparison concerns large positive $\ell$, not $\ell = 0$.

Source: V-B; convergence paragraph; SRC-024.

**T2.3.** $b_{\ell-1}/b_\ell = \alpha^{-1}((2\ell+1)/(2\ell-1))^{D-1}F_{\ell-1}/F_\ell \to \alpha^{-1} < 1$.

*Justification.* Use the factorization in T2.1, the positive finite limit from T2.2, and elementary limits.

Source: V-B; ratio display; SRC-025.

**T2.4.** There are $q \in (0,1)$ and $L$ with $b_{\ell-1} \le qb_\ell$ for $\ell \ge L$, and $b_{N-1} \le Q_D(N) \le C'b_{N-1}$ for all sufficiently large $N$.

*Justification.* The ratio limit supplies $q, L$. The source bounds the tail by $b_{N-1}/(1-q)$ and absorbs the fixed prefix $\sum_{\ell=0}^{L-1} b_\ell$ into $C'b_{N-1}$. Eventual growth and the qualification "sufficiently large are implicit in that last absorption; see A-02.

Source: V-B; geometric domination display; SRC-026.

**T2.5.** Q.E.D.: $Q_D(N) = \Theta(\alpha^N / N^{D-1})$.

*Justification.* $b_{N-1} = \alpha^N F_{N-1}/(2N-1)^{D-1}$ has this order by T2.2. Combine with T2.4. Constants may depend on fixed $D, p$.

Source: V-B; final sentence; SRC-027.

## 4.6 Remark 3: sharper leading constant

**ASSUME / PROVE.** Under the assumptions of Theorem 2, the paper claims $Q_D(N) \sim \frac{\alpha}{\alpha-1} F_\infty \frac{\alpha^N}{(2N)^{D-1}}$.

**Source:** V-B; Eq. (25). **Forward:** PASS. **Reverse:** FOLLOWS.

**R3.1.** Put $r = \alpha^{-1} \in (0,1)$. For each fixed integer $j \geq 0$, $b_{N-1-j}/b_{N-1} \to r^j$.

*Justification.* For $N > j$, express the ratio as the product of $j$ consecutive backward ratios. Each tends to $r$ by Theorem 2. The empty product at $j = 0$ is one.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-028.

**R3.2.** Choose $r < q < 1$ and $L$ so that $b_{\ell-1} \leq q b_\ell$ for all $\ell \geq L$. For $N - 1 - j \geq L$, $b_{N-1-j}/b_{N-1} \leq q^j$.

*Justification.* The backward ratio limit gives this threshold. Repeatedly apply the inequality through the finite product of ratios. All $b_\ell$ are positive.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-029.

**R3.3.** $Q_D(N)/b_{N-1} \to \sum_{j=0}^{\infty} r^j = \alpha/(\alpha-1)$.

*Justification.* Read the sum from $N-1$ backward. Set terms outside $0 \leq j \leq N-1-L$ to zero. For each fixed $j$, they eventually have the limit in R3.1; R3.2 bounds them by the summable sequence $q^j$. Dominated convergence for a series gives the tail limit. The fixed prefix $\sum_{\ell<L} b_\ell/b_{N-1}$ vanishes because $b_{N-1} \to \infty$, as supplied by the factorization and $\alpha > 1$.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-030.

**R3.4.** $Q_D(N) \sim \frac{\alpha}{\alpha-1} F_\infty \frac{\alpha^N}{(2N)^{D-1}}$.

*Justification.* Use $b_{N-1} = \alpha^N F_{N-1}/(2N-1)^{D-1}$, $F_{N-1} \to F_\infty > 0$, and $(2N-1)/(2N) \to 1$, with fixed $D$. Combine with R3.3.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-031.

**R3.5.** Q.E.D. for Remark 3.

*Justification.* R3.4 is the claimed sharper equivalent. The former GAP-001 is closed by the explicit added source argument.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-032.

## 4.7 Lemma 1: hitting time via capacity

**ASSUME / PROVE.** Fix $D \geq 1$, $0 < p < 1$, a nonempty finite $A \subset \mathbb{Z}^D$, and $a \notin A$. For the un-killed chain, put $h(x) = \Pr_x(\tau_a < \tau_A)$ and $M = \sum_x \mu(x)$. The claims are

$$\mathbb{E}_a \tau_A = \frac{\sum_x \mu(x) h(x)}{\mathrm{Cap}(a, A)}, \qquad \mu(a) R_{\mathrm{eff}}(a, A) \leq \mathbb{E}_a \tau_A \leq M R_{\mathrm{eff}}(a, A).$$

**Source:** V-C; Eqs. (26)–(27). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

**L1.1.** $M = 1+\sum_{k\geq 1} S_D(k)p^k < \infty$; the irreducible chain has stationary probability $\mu/M$ and is positive recurrent. The source concludes that the fixed-state hitting and return times used below have finite means.

*Justification.* **EXT-001; EXTERNAL UNAVAILABLE**. The mass series converges by polynomial–geometric summability. The invoked countable-chain recurrence and hitting-time background is retained as EXT-001; the exact external theorem statement is not supplied.

Source: V-C; proof opening; SRC-033.

**L1.O1.** OBLIGATION: obtain the exact countable-state recurrence/hitting-time result invoked by L1.1 and verify it for this irreducible chain with invariant probability $\mu/M$.

*Justification.* **EXT-001; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: V-C; proof opening; SRC-034.

**L1.2.** For $h(x) = \Pr_x(\tau_a < \tau_A)$, $h(a) = 1$, $h|_A = 0$, and $Ph = h$ off $\{a\} \cup A$. For bounded functions,

$$\mathcal{E}(f, g) = \sum_x \mu(x)g(x)(f(x) - Pf(x)).$$

*Justification.* Condition on the first step for harmonicity. Expand the symmetric mixed energy and use detailed balance. Boundedness and $\sum_{x,y} \mu(x)P(x, y) = M < \infty$ justify absolute convergence. Self-loops give zero energy.

Source: V-C; harmonicity and mixed energy identity; SRC-035.

**L1.3.** With $q = \Pr_a(\tau_A < \tau_a^+)$, $\mathcal{E}(h, h) = \mu(a)(1 - Ph(a)) = \mu(a)q$.

*Justification.* In L1.2, only the contribution at $a$ survives. The first-step return decomposition gives $1 - Ph(a) = q$. Recurrence excludes avoiding both boundary sets forever. Rejected proposals count as returns at time one.

Source: V-C; capacity computation; SRC-036.

**L1.4.** $h$ minimizes energy over $f(a) = 1$, $f|_A = 0$.

*Justification.* The source first clips $f$ into $[0, 1]$, which preserves its boundary values and cannot increase energy. Write the clipped $f = h + g$. Harmonicity and $g|_{\{a\}\cup A} = 0$ give $\mathcal{E}(h, g) = 0$. Hence $\mathcal{E}(f, f) = \mathcal{E}(h, h) + \mathcal{E}(g, g) \geq \mathcal{E}(h, h)$. All functions in this expansion are bounded.

Source: V-C; energy minimization; SRC-037.

**L1.5.** $\mathrm{Cap}(a, A) = \mu(a)q$.

*Justification.* Use the attained variational minimum in L1.4 and its value in L1.3. A simple finite path from $a$ into $A$, without a prior return to $a$, has positive probability, so $q > 0$. The positivity check is implicit in irreducibility.

Source: V-C; end of capacity argument; SRC-038.

**L1.6.** For $G_A(u, x) = \mathbb{E}_u \sum_{t<\tau_A} \mathbf{1}_{\{X_t=x\}}$,

$$\mathbb{E}_a\tau_A = \sum_x G_A(a, x), \qquad \mu(a)G_A(a, x) = \mu(x)G_A(x, a).$$

*Justification.* Sum the nonnegative occupation counts. Apply detailed balance to each finite path avoiding $A$, reverse the path, and sum over lengths. This is reversibility of the killed Green function, not of an absorbing transition kernel with positive target weights.

Source: V-C; occupation sum and killed reversibility; SRC-039.

**L1.7.** $G_A(x, a) = h(x)G_A(a, a)$ and $G_A(a, a) = 1/q = \mu(a)/\mathrm{Cap}(a, A)$.

*Justification.* Use the strong Markov property at the first hit of $a$ before $A$. Starting at $a$, each visit has probability $q$ of escape before the next return; the number of visits, including the initial one, is geometric. This argument is explicit in the latest source.

Source: V-C; strong Markov property and geometric visits; SRC-040.

**L1.8.** $\mathbb{E}_a \tau_A = \big(\sum_x \mu(x)h(x)\big) / \operatorname{Cap}(a, A)$.

*Justification.* Substitute the two Green identities from L1.6–L1.7 into the occupation sum and cancel the positive $\mu(a)$.

Source: V-C; Eq. (26); SRC-041.

**L1.9.** $R_{\text{eff}}(a, A) = 1/\operatorname{Cap}(a, A)$, so $\mu(a)R_{\text{eff}} \le \mathbb{E}_a \tau_A \le M R_{\text{eff}}$.

*Justification.* **EXT-002; EXTERNAL UNAVAILABLE**. The source now names Dirichlet's principle: the potential-energy minimum is effective conductance, whose reciprocal is resistance. The exact countably infinite finite-target statement remains EXT-002. Bound the numerator in L1.8 using $h(a) = 1$ and $0 \le h \le 1$.

Source: V-C; Eqs. (26)–(27); SRC-042.

**L1.O2.** OBLIGATION: obtain and check the exact finite-target, countably infinite network Dirichlet capacity/resistance identity and the Thomson unit-flow inequality, matching the source conventions and hypotheses.

*Justification.* **EXT-002; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: V-C; capacity convention; V-E; Thomson application; SRC-043.

**L1.10.** Q.E.D. for Lemma 1, conditional on EXT-001 and EXT-002.

*Justification.* L1.8 proves the identity; L1.9 gives both inequalities.

Source: V-C; conclusion; SRC-044.

## 4.8 Theorem 3: true-chain lower bound

**ASSUME / PROVE.** For fixed integer $D \ge 1$, $0 < p < 1$, and every integer $N \ge 1$,

$$H_D(N) \ge \frac{\alpha^N}{(2N-1)^{D-1}} \ge 2^{-(D-1)} \frac{\alpha^N}{N^{D-1}}.$$

**Source:** V-D; Eq. (28). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

**T3.1.** The exterior of $B_N$ does not change the resistance from 0 to $A_N$.

*Justification.* The shell separates 0 from the exterior; an equilibrium potential is zero outside. Thus shorting may be performed on the finite cube. This is the source restriction argument.

Source: V-D; separation argument; SRC-045.

**T3.2.** $R_{\text{eff}}(0, A_N) \ge R_{\text{eff}}^{\text{coll}}(0, N)$.

*Justification.* Short all vertices of each shell. Rayleigh's shorting principle applies to this finite network with positive conductances; the source cites Doyle–Snell and Levin–Peres.

Source: V-D; Rayleigh monotonicity; SRC-046.

**T3.3.** $R_{\text{eff}}^{\text{coll}}(0, N) = \sum_{k=0}^{N-1} \alpha^{k+1}/(2k+1)^{D-1}$, of order $\alpha^N / N^{D-1}$.

*Justification.* The intershell conductances from P1 are in series. The ratio/geometric-tail calculation is the one used in Theorem 2 with the factor $F_\ell$ absent. The next source step supplies an alternative lower estimate requiring no asymptotic summation.

Source: V-D; resistance sum; SRC-047.

**T3.4.** $H_D(N) \geq R_{\text{eff}}(0, A_N) \geq \alpha^N/(2N-1)^{D-1}$.

*Justification.* Apply the lower inequality of Lemma 1 with $a = 0$, where $\mu(0) = 1$. In the positive series of T3.3 retain only $k = N - 1$.

Source: V-D; last-term refinement; SRC-048.

**T3.5.** Q.E.D. with $c_{D,p} = 2^{-(D-1)}$.

*Justification.* Since $2N - 1 \leq 2N$ and $D - 1 \geq 0$, T3.4 implies the asserted inequality for every $N \geq 1$. Its dependence on the unresolved background of L1 remains explicit.

Source: V-D; last display; SRC-049.

## 4.9 Lemma 2: transport on nested cubes

> **ASSUME / PROVE.** For integers $n \geq 1$, $k \geq 0$, let $Q_k = \{-k, \ldots, k\}^n$, $\rho_k$ be its one-dimensional uniform law extended by zero, and $u_k = \rho_k^{\otimes n}$. There is a nearest-neighbor flow $\psi_k$ on $Q_{k+1}$ such that $\operatorname{div} \psi_k = u_k - u_{k+1}$ and $\sum_e \psi_k(e)^2 \leq n/(k+1)^n$, counting each unoriented edge once.

**Source:** V-E; Eq. (29). **Forward:** PASS. **Reverse:** FOLLOWS.

**L2.1.** For $n = 1$, orient edges $j \to j + 1$ and set $J_{j+1/2} = \sum_{t=-k-1}^{j} (\rho_k(t) - \rho_{k+1}(t))$ for $-k - 1 \leq j \leq k$, with flux zero outside.

*Justification.* This is the source definition. It specifies signed flow on every oriented edge, with the reverse orientation its negative.

Source: V-E; one-dimensional construction; SRC-050.

**L2.2.** $J_{j+1/2} - J_{j-1/2} = \rho_k(j) - \rho_{k+1}(j)$, including the endpoints, and

$$J_{j+1/2} = -\frac{1}{2k+3} + \frac{2(j+k+1)}{(2k+1)(2k+3)}.$$

*Justification.* Telescope the cumulative sums for divergence. For the displayed flux, count the interior terms in the sum of L2.1. The endpoint convention uses that the total difference of the two probability masses is zero.

Source: V-E; divergence and explicit evaluation; SRC-051.

**L2.3.** $|J_{j+1/2}| \leq 1/(2k+3)$ and $\sum_{j=-k-1}^{k} J_{j+1/2}^2 \leq (2k+2)/(2k+3)^2 \leq 1/(k+1)$.

*Justification.* The explicit affine formula in L2.2 ranges between the two endpoint values $-1/(2k+3)$ and $1/(2k+3)$. There are $2k + 2$ edges. All denominators are positive, including at $k = 0$.

Source: V-E; one-dimensional energy bound; SRC-052.

**L2.4.** Define $\nu^{(r)} = \rho_{k+1}^{\otimes r} \otimes \rho_k^{\otimes(n-r)}$, $0 \leq r \leq n$. For stage $r < n$, the flow in coordinate $r + 1$ is $w_r(z) J_{j+1/2}$, where $w_r(z) = \prod_{i=1}^{r} \rho_{k+1}(z_i) \prod_{i=r+2}^{n} \rho_k(z_i)$.

*Justification.* The source changes one coordinate at a time. Other directions have zero stage flow. Empty products equal one.

Source: V-E; product-stage construction; SRC-053.

**L2.5.** $\operatorname{div} \psi^{(r)} = \nu^{(r)} - \nu^{(r+1)}$.

*Justification.* On a fixed fiber, multiply the divergence identity L2.2 by the constant weight $w_r(z)$. The product laws differ only in the active coordinate.

Source: V-E; fiber divergence; SRC-054.

**L2.6.** $\sum_z w_r(z)^2 = (2k+3)^{-r}(2k+1)^{-(n-r-1)} \le (k+1)^{-(n-1)}$.

*Justification.* Factor the finite sum of products. A uniform law on $m$ points has sum of squares $1/m$. There are $n-1$ fixed coordinates; $2k+1, 2k+3 \ge k+1$.

Source: V-E; squared fiber masses; SRC-055.

**L2.7.** $\sum_e \psi^{(r)}(e)^2 = (\sum_z w_r(z)^2)(\sum_j J_{j+1/2}^2) \le (k+1)^{-n}$.

*Justification.* The fiber index and edge index separate, so use L2.3 and L2.6.

Source: V-E; stage energy; SRC-056.

**L2.8.** For $\psi_k = \sum_{r=0}^{n-1} \psi^{(r)}$, the divergence is $u_k - u_{k+1}$ and the energy is at most $n/(k+1)^n$.

*Justification.* The divergences in L2.5 telescope between $\nu^{(0)} = u_k$ and $\nu^{(n)} = u_{k+1}$. Distinct stages have distinct coordinate directions and therefore disjoint unoriented edge sets. Their squared energies add without cross terms.

Source: V-E; summing stage flows; SRC-057.

**L2.9.** Q.E.D. with $C_n = n$, including $k = 0$.

*Justification.* Use the explicitly constructed flow L2.8; no limiting argument or external network theorem is required.

Source: V-E; end of transport proof; SRC-058.

## 4.10 Proposition 2: resistance upper bound

> **ASSUME / PROVE.** For fixed $D \ge 1$, $0 < p < 1$, let $F_N^\circ = \{(N, y) : |y_i| \le N-1\}$. There is $C'_{D,p} < \infty$, independent of integer $N \ge 1$, with $R_{\text{eff}}(0, F_N^\circ) \le C'_{D,p}\alpha^N/N^{D-1}$.

**Source:** V-E; Eq. (30). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

**P2.1.** CASE $D = 1$: unit current on the positive path has energy $2\sum_{k=0}^{N-1} \alpha^{k+1} \le 2\alpha^{N+1}/(\alpha-1)$.

*Justification.* Each path edge has conductance $p^{k+1}/2$. Thomson's inequality gives the resistance upper bound. The factor two from the revised source is retained. The infinite-network application is covered by EXT-002.

Source: V-E; D=1 correction; SRC-059.

**P2.2.** CASE $D \ge 2$: put $n = D-1$; use layers $L_k = \{(k, y) : y \in Q_k\}$, $0 \le k < N$, and target $L_N^\circ = \{(N, y) : y \in Q_{N-1}\}$.

*Justification.* The source builds the following unit flow. At each departure layer $L_k$ the outgoing distribution is uniform on $Q_k$.

Source: V-E; D>=2 construction; SRC-060.

> **P2.2.1.** Each of the $(2k+1)^n$ vertical edges carries $(2k+1)^{-n}$, with energy $E_k^{\text{vert}} = 2D\alpha^{k+1}/(2k+1)^n \le 2D\alpha^{k+1}/(k+1)^n$.
>
> *Justification.* For such edges the energy increases from $k$ to $k+1$, so the conductance is $p^{k+1}/(2D)$. Multiply the squared current divided by conductance by the number of edges.
>
> Source: V-E; forward step; SRC-061.

> **P2.2.2.** For $0 \le k \le N-2$, redistribute $u_k$ to $u_{k+1}$ within $L_{k+1}$ at energy $E_{k+1}^{\text{hor}} \le 2Dn\alpha^{k+1}/(k+1)^n$.
>
> *Justification.* Apply Lemma 2 and multiply its unit-conductance cost by $2D\alpha^{k+1}$. All vertices of this layer have energy $k+1$, so each lateral conductance is $p^{k+1}/(2D)$.
>
> Source: V-E; lateral redistribution; SRC-062.

**P2.2.3.** At intermediate $L_j$, divergence is $-u_{j-1} + u_j + (u_{j-1} - u_j) = 0$; source divergence is 1 and total target divergence is $-1$.

*Justification.* Incoming vertical current contributes the negative term; outgoing vertical current the positive term; Lemma 2 gives lateral divergence. Extend $u_{j-1}$ by zero outside $Q_{j-1}$. At the last step there is no redistribution. At $N = 1$ there is only one vertical edge.

Source: V-E; Conservation of current; SRC-063.

**P2.2.4.** After extension by zero, the flow is feasible on $\mathbb{Z}^D$ and

$$R_{\text{eff}}(0, F_N^\circ) \le \sum_{k=0}^{N-1} E_k^{\text{vert}} + \sum_{k=1}^{N-1} E_k^{\text{hor}} \le C_{D,p} \sum_{k=0}^{N-1} \frac{\alpha^{k+1}}{(k+1)^n}.$$

*Justification.* Vertical edges and lateral edges in distinct layers are disjoint. Add their energies and apply Thomson's principle, subject to EXT-002.

Source: V-E; energy sum and Thomson; SRC-064.

**P2.2.5.** For $a_j = \alpha^j / j^n$, choose $q \in (\alpha^{-1}, 1)$ and $J$ with $a_{j-1} \le q a_j$ for $j \ge J + 1$. Then $\sum_{j=1}^N a_j \le K_{n,p} a_N$ for all $N \ge 1$.

*Justification.* The ratio tends to $\alpha^{-1}$. For $N \ge J$, bound the tail by $a_N/(1-q)$ and absorb the finite prefix using $a_N \ge a_J > 0$. Enlarge the constant for the finitely many $1 \le N < J$. This entire argument is present in the revised source.

Source: V-E; ratio argument; SRC-065.

**P2.2.6.** Q.E.D. for the $D \ge 2$ case, conditional on EXT-002.

*Justification.* Combine P2.2.4 with P2.2.5 and $n = D - 1$. Export the completed case as P2.2.

Source: V-E; end of D>=2 case; SRC-066.

**P2.3.** Q.E.D. for Proposition 2.

*Justification.* The cases $D = 1$ and $D \ge 2$ exhaust integer $D \ge 1$. Use the completed cases P2.1 and P2.2. The constant depends on $D, p$, not $N$.

Source: V-E; end of proposition proof; SRC-067.

## 4.11 Theorem 4: true-chain asymptotic scaling

**ASSUME / PROVE.** For fixed integer $D \ge 1$, fixed $0 < p < 1$, and the true Metropolis chain started at 0, $H_D(N) = \Theta(\alpha^N / N^{D-1})$ as $N \to \infty$.

**Source:** V-E; Eq. (31). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

**T4.1.** $H_D(N) \ge 2^{-(D-1)} \alpha^N / N^{D-1}$.

*Justification.* Invoke Theorem 3.

Source: Theorem 4 proof; opening; SRC-068.

**T4.2.** $H_D(N) \le \mathbb{E}_0 \tau_{F_N^\circ} \le M R_{\text{eff}}(0, F_N^\circ) \le M C'_{D,p} \alpha^N / N^{D-1}$.

*Justification.* $F_N^\circ \subset A_N$ gives the pathwise comparison of first hitting times. Apply Lemma 1 to the nonempty finite face, then Proposition 2.

Source: Theorem 4 proof; upper display; SRC-069.

**T4.3.** Q.E.D. for the claimed $\Theta$ law, conditional on the recorded background obligations.

*Justification.* The finite $M = M_{D,p}$ and the constants in T4.1–T4.2 depend only on fixed $D, p$. The positive upper and lower constants establish the claimed order.

Source: Theorem 4 proof; final sentence; SRC-070.

## 4.12 Ancillary symmetry and numerical-method identities

> **ASSUME / PROVE.** The shell quotient is generally not the exact shell dynamics, whereas grouping points by coordinate signs and order preserves the true dynamics. The active Section VI uses sparse escape-probability and excursion-time systems. The comparison to the earlier one-dimensional paper is separate external support.

**Source:** II; III-C; IV-C Remark 1; VI-A. **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

**N.1.** For $D \geq 2$, $1 \leq k < N$, $(k, 0, \ldots, 0)$ and $(k, k, 0, \ldots, 0)$ have outward proposal probabilities $1/(2D)$ and $1/D$.

*Justification.* Count the coordinates on the shell boundary. The accepted outward probabilities are $p/(2D)$ and $p/D$. Thus shell membership alone does not specify the transition probabilities; the shell quotient is a separate chain.

Source: III-C; shell counterexample; SRC-071.

**N.2.** Grouping lattice points that differ only in coordinate signs or coordinate order preserves the transition probabilities between groups and the escape-time calculation.

*Justification.* **EXT-003; EXTERNAL UNAVAILABLE**. The source invokes the Kemeny–Snell aggregation criterion and the kernel symmetry. EXT-003 records the exact external-text check. No group-theory notation is required.

Source: II; aggregation paragraph; VI-A; SRC-072.

**N.O3.** OBLIGATION: obtain Kemeny–Snell Sec. 6.3, Thm. 6.3.2 and verify its aggregation criterion for the stated symmetry groups.

*Justification.* **EXT-003; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: II; aggregation paragraph; SRC-073.

**N.3.** The true mean is computed using sparse systems for escape probabilities and excursion times on these symmetry-equivalent groups.

*Justification.* This is the active numerical-method description. The earlier explicit regenerative formula is now commented out in the source; it is not treated as a current displayed claim. Numerical accuracy and plotted speedups are outside this proof audit.

Source: VI-A; Numerical Method; SRC-074.

**N.4.** The $D = 1$ quotient formula matches the predecessor after the parameter identification $p = 2p_{\text{prior}}$.

*Justification.* **EXT-004; EXTERNAL UNAVAILABLE**. The source cites its predecessor. The predecessor's exact parameter convention and formula were not included in this audit (EXT-004). This comparison is not used to establish Theorems 1–4.

Source: IV-C; Remark 1; SRC-075.

**N.O4.** OBLIGATION: obtain the predecessor formula and parameter convention, and check $p = 2p_{\text{prior}}$.

*Justification.* **EXT-004; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: IV-C; Remark 1; SRC-076.

# 5 Source-to-step mapping ledger

Each contract maps to the source unit identified in the inventory. The rows below map every rendered step. Source IDs index the precise paraphrases printed at those steps; they are provenance, not extra premises. A repeated source paragraph may produce several normalized steps. Obligation rows preserve missing support and do not count as supplied proof evidence.

| Step | Source ID | Mapping | Support | Issue | Notes |
|---|---|---|---|---|---|
| G contract | SRC-G-C | NORMALIZED | EXPLICIT | – | Contract |
| G.1 | SRC-001 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.2 | SRC-002 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.3 | SRC-003 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.3.1 | SRC-004 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.3.2 | SRC-005 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.4 | SRC-006 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.5 | SRC-007 | NORMALIZED | EXPLICIT | – | Same source location as step |
| G.6 | SRC-008 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| T1 contract | SRC-T1-C | NORMALIZED | EXPLICIT | – | Contract |
| T1.1 | SRC-009 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T1.2 | SRC-010 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T1.3 | SRC-011 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T1.4 | SRC-012 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T1.5 | SRC-013 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T1.6 | SRC-014 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| S contract | SRC-S-C | NORMALIZED | EXPLICIT | – | Contract |
| S.1 | SRC-015 | NORMALIZED | EXPLICIT | – | Same source location as step |
| S.2 | SRC-016 | NORMALIZED | EXPLICIT | – | Same source location as step |
| S.3 | SRC-017 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| P1 contract | SRC-P1-C | NORMALIZED | EXPLICIT | – | Contract |
| P1.1 | SRC-018 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P1.2 | SRC-019 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P1.3 | SRC-020 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P1.4 | SRC-021 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P1.5 | SRC-022 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| T2 contract | SRC-T2-C | NORMALIZED | EXPLICIT | – | Contract |
| T2.1 | SRC-023 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T2.2 | SRC-024 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T2.3 | SRC-025 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T2.4 | SRC-026 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T2.5 | SRC-027 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| R3 contract | SRC-R3-C | NORMALIZED | EXPLICIT | – | Contract |

| Step | Source ID | Mapping | Support | Issue | Notes |
|---|---|---|---|---|---|
| R3.1 | SRC-028 | NORMALIZED | EXPLICIT | – | Same source location as step |
| R3.2 | SRC-029 | NORMALIZED | EXPLICIT | – | Same source location as step |
| R3.3 | SRC-030 | NORMALIZED | EXPLICIT | – | Same source location as step |
| R3.4 | SRC-031 | NORMALIZED | EXPLICIT | – | Same source location as step |
| R3.5 | SRC-032 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| L1 contract | SRC-L1-C | NORMALIZED | EXPLICIT | – | Contract |
| L1.1 | SRC-033 | NORMALIZED | EXTERNAL UNAVAILABLE | EXT-001 | Same source location as step |
| L1.O1 | SRC-034 | OBLIGATION | EXTERNAL UNAVAILABLE | EXT-001 | Same source location as step |
| L1.2 | SRC-035 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.3 | SRC-036 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.4 | SRC-037 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.5 | SRC-038 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.6 | SRC-039 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.7 | SRC-040 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.8 | SRC-041 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L1.9 | SRC-042 | NORMALIZED | EXTERNAL UNAVAILABLE | EXT-002 | Same source location as step |
| L1.O2 | SRC-043 | OBLIGATION | EXTERNAL UNAVAILABLE | EXT-002 | Same source location as step |
| L1.10 | SRC-044 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| T3 contract | SRC-T3-C | NORMALIZED | EXPLICIT | – | Contract |
| T3.1 | SRC-045 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T3.2 | SRC-046 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T3.3 | SRC-047 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T3.4 | SRC-048 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T3.5 | SRC-049 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| L2 contract | SRC-L2-C | NORMALIZED | EXPLICIT | – | Contract |
| L2.1 | SRC-050 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.2 | SRC-051 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.3 | SRC-052 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.4 | SRC-053 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.5 | SRC-054 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.6 | SRC-055 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.7 | SRC-056 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.8 | SRC-057 | NORMALIZED | EXPLICIT | – | Same source location as step |
| L2.9 | SRC-058 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| P2 contract | SRC-P2-C | NORMALIZED | EXPLICIT | – | Contract |
| P2.1 | SRC-059 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2 | SRC-060 | NORMALIZED | EXPLICIT | – | Same source location as step |

| Step | Source ID | Mapping | Support | Issue | Notes |
|---|---|---|---|---|---|
| P2.2.1 | SRC-061 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2.2 | SRC-062 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2.3 | SRC-063 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2.4 | SRC-064 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2.5 | SRC-065 | NORMALIZED | EXPLICIT | – | Same source location as step |
| P2.2.6 | SRC-066 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| P2.3 | SRC-067 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| T4 contract | SRC-T4-C | NORMALIZED | EXPLICIT | – | Contract |
| T4.1 | SRC-068 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T4.2 | SRC-069 | NORMALIZED | EXPLICIT | – | Same source location as step |
| T4.3 | SRC-070 | STRUCTURAL | IMPLICIT | – | Same source location as step |
| N contract | SRC-N-C | NORMALIZED | EXPLICIT | – | Contract |
| N.1 | SRC-071 | NORMALIZED | EXPLICIT | – | Same source location as step |
| N.2 | SRC-072 | NORMALIZED | EXTERNAL UNAVAILABLE | EXT-003 | Same source location as step |
| N.O3 | SRC-073 | OBLIGATION | EXTERNAL UNAVAILABLE | EXT-003 | Same source location as step |
| N.3 | SRC-074 | NORMALIZED | EXPLICIT | – | Same source location as step |
| N.4 | SRC-075 | NORMALIZED | EXTERNAL UNAVAILABLE | EXT-004 | Same source location as step |
| N.O4 | SRC-076 | OBLIGATION | EXTERNAL UNAVAILABLE | EXT-004 | Same source location as step |

# 6 Gap and external-support register

**GAP-001 — CLOSED in this revision.** Remark 3 now proves

$$Q_D(N)/b_{N-1} \longrightarrow \alpha/(\alpha-1).$$

Steps R3.1–R3.3 give pointwise limits and geometric domination of the backward terms, together with disappearance of the fixed prefix. R3.4 converts this limit to the stated sharper equivalent. This closure applies to the quotient refinement and does not change the dependency route for the main true-chain theorem.

**EXT-001 — countable-state background.** Obtain the exact invoked theorem statements and check that irreducibility plus the invariant probability $\mu/M$ supplies recurrence and the fixed-state finite hitting/return means used in L1.1. The source's polynomial–geometric mass estimate is checked. Unavailable support propagates through L1, T3 and T4. The explicit meta-obligation is L1.O1.

**EXT-002 — finite-target infinite-network convention.** Obtain and check the exact capacity/resistance duality and the unit-flow inequality for the countable network with the variational capacity defined above. Match finite targets, boundary behavior, normalization and self-loops. L1.O2 records this obligation; it affects L1.9, the network application in P2, and T3/T4. The finite-support flow and its energy are locally checked independently of this external boundary.

**EXT-003 — aggregation criterion.** Obtain Kemeny–Snell Sec. 6.3, Thm. 6.3.2 and verify its exact applicability to the grouping by coordinate signs/order. N.O3 records this obligation; it affects N.2–N.3, not the main analytical theorem.

**EXT-004 — predecessor comparison.** Obtain the cited predecessor's exact formula and proposal/acceptance convention to verify $p = 2p_{\text{prior}}$. N.O4 records the obligation; only N.4 depends on it.

**Editorial changes applied.** Equation `eq:sharp-asymptotic` now has ordinary multiplication spacing and the

full displayed factor. The bias-ratio definition permits $\alpha = 1$, while its strict inequality is restricted to $p < 1$. The repository paragraph names the numerical notebook and Lamport-proof supplement; the authors will upload the latter. The revised acknowledgment names ChatGPT, Claude and Grok and retains author responsibility.

## 7 Frozen audit handoff

The contracts, rendering, source identity, step labels, support statuses and register above form the frozen handoff. The audits below evaluate that rendering with GAP-001 explicitly closed by new source text, while unavailable external support remains open. Mapping metadata is not proof evidence. Any future correction should change the source hash and identify which obligations it closes.

# Part II

# Forward audit

## 8 Verdict and theorem contracts

**Overall verdict: INCOMPLETE under the frozen external-source boundary.** The previous internal gap in Remark 3 is closed. The exact quotient formula, its asymptotic order, the sharper equivalent, and the transport construction close with at most minor exposition clarifications. The true-chain proof still invokes the recorded countable-state and network results whose exact external statements were not reverified here. The four theorem contracts are T1, T2, T3 and T4; limits keep $D, p$ fixed.

## 9 Proof structure and findings

The rendering contains twelve units and two numbered levels within the constructive proofs. Source order is preserved within each mathematical argument. Lemma 1 now proves the capacity identity before using it in the Green-function hitting-time formula; the previous supplement's provisional order and descriptions have been updated.

**RESOLVED — R3.1–R3.5.** The new source gives the formerly missing leading-constant bridge. Each backward ratio tends to $r = \alpha^{-1}$; eventual geometric domination supplies a summable majorant; the fixed prefix vanishes. The resulting normalized sum tends to $1/(1 - r)$. No limit/sum exchange is asserted without its domination condition.

**CONDITIONAL — L1.1, L1.9 and P2.** Exact external applicability remains EXT-001/002. The paper's latest capacity minimization and Green-function argument are substantially more explicit than in the old supplement. This resolves the earlier compressed local exposition, but does not by itself verify an external theorem's full scope.

**MINOR — T1.1 and T1.3.** The finite quotient has finite mean absorption: a consecutive outward path has positive probability from every transient state; finitely many states give a uniform positive chance of absorption in a bounded block. Geometric survival bounds justify the first-step mean equations. Empty products equal one. These elementary checks are audit clarifications, not extra source proof text.

**MINOR — T2.4.** The geometric-tail comparison holds for sufficiently large $N$. The fixed initial prefix is absorbed into a constant times the eventually increasing positive final term. Proposition 2 already spells out its analogous finite-prefix argument.

**Resolved local clarification — L1.4/L1.7.** The latest source clips the competing function before expanding the energy and explicitly derives geometric visit counts. The older suggestion to add these explanations should no longer be reported as pending.

**NOTE — symmetry and numerics.** The latest active Section VI uses plain grouping terminology and omits the old displayed renewal formula. The formula survives only in comments and is not converted into an active

current claim. Empirical accuracy statements do not support any proof step.

**Scope discipline.** A completed parent step, rather than its private descendants, supplies later unit dependencies. The local transport construction and dimension cases have explicit closure. No circularity or use of the unproved pointwise inequality $Q_D \le H_D$ was found. Every limit, denominator and energy bound is interpreted in its declared domain.

# 10 Step ledger

The assertion and checked justification for every row appear in the rendering. The current goal is the enclosing unit contract, or a nested parent claim for its children. The contracts supply $D, N, p$ and local domains; definitions and case assumptions remain local. Obligation rows are not successful proof steps.

| Step | Goal | Legal dependencies / justification location | Status |
|---|---|---|---|
| G.1 | G | contract; checked text at G.1 | OK |
| G.2 | G | G.1; checked text at G.2 | OK |
| G.3 | G | G.1, G.2; checked text at G.3 | OK |
| G.3.1 | G.3 | G.1, binomial theorem; checked text at G.3.1 | OK |
| G.3.2 | G.3 | G.3.1, G.2; checked text at G.3.2 | OK |
| G.4 | G | G.1, G.2; checked text at G.4 | OK |
| G.5 | G | G.3, G.4; checked text at G.5 | OK |
| G.6 | G | G.2, G.3, G.4, G.5; checked text at G.6 | OK |
| T1.1 | T1 | G; checked text at T1.1 | MINOR |
| T1.2 | T1 | T1.1, G, G; checked text at T1.2 | OK |
| T1.3 | T1 | T1.2; checked text at T1.3 | MINOR |
| T1.4 | T1 | T1.1, T1.2, T1.3; checked text at T1.4 | OK |
| T1.5 | T1 | T1.4; checked text at T1.5 | OK |
| T1.6 | T1 | T1.1, T1.5; checked text at T1.6 | OK |
| S.1 | S | G, T1; checked text at S.1 | OK |
| S.2 | S | G, T1; checked text at S.2 | OK |
| S.3 | S | S.2; checked text at S.3 | OK |
| P1.1 | P1 | contract; checked text at P1.1 | OK |
| P1.2 | P1 | P1.1, G, G; checked text at P1.2 | OK |
| P1.3 | P1 | P1.2, G, G; checked text at P1.3 | OK |
| P1.4 | P1 | P1.1; checked text at P1.4 | OK |
| P1.5 | P1 | P1.1, P1.3, P1.4; checked text at P1.5 | OK |
| T2.1 | T2 | T1; checked text at T2.1 | OK |
| T2.2 | T2 | G, T2.1; checked text at T2.2 | OK |
| T2.3 | T2 | T2.1, T2.2; checked text at T2.3 | OK |
| T2.4 | T2 | T1, T2.3; checked text at T2.4 | MINOR |
| T2.5 | T2 | T2.2, T2.4; checked text at T2.5 | OK |
| R3.1 | R3 | T2; checked text at R3.1 | OK |
| R3.2 | R3 | T2, R3.1; checked text at R3.2 | OK |
| R3.3 | R3 | T1, T2, R3.1, R3.2; checked text at R3.3 | OK |
| R3.4 | R3 | T2, R3.3; checked text at R3.4 | OK |
| R3.5 | R3 | R3.4; checked text at R3.5 | OK |
| L1.1 | L1 | G, P1; checked text at L1.1 | CONDITIONAL |
| L1.O1 | L1 | source citation; checked text at L1.O1 | CONDITIONAL |
| L1.2 | L1 | L1.1, P1; checked text at L1.2 | CONDITIONAL |
| L1.3 | L1 | L1.1, L1.2; checked text at L1.3 | CONDITIONAL |
| L1.4 | L1 | L1.2, L1.3; checked text at L1.4 | CONDITIONAL |
| L1.5 | L1 | L1.3, L1.4; checked text at L1.5 | CONDITIONAL |
| L1.6 | L1 | L1.1, P1; checked text at L1.6 | CONDITIONAL |
| L1.7 | L1 | L1.1, L1.5, L1.6; checked text at L1.7 | CONDITIONAL |
| L1.8 | L1 | L1.6, L1.7; checked text at L1.8 | CONDITIONAL |
| L1.9 | L1 | L1.8, L1.1; checked text at L1.9 | CONDITIONAL |
| L1.O2 | L1 | source citation; checked text at L1.O2 | CONDITIONAL |
| L1.10 | L1 | L1.8, L1.9; checked text at L1.10 | CONDITIONAL |
| T3.1 | T3 | L1; checked text at T3.1 | OK |
| T3.2 | T3 | T3.1, P1; checked text at T3.2 | OK |
| T3.3 | T3 | P1, T2, T2; checked text at T3.3 | OK |
| T3.4 | T3 | L1, T3.2, T3.3; checked text at T3.4 | CONDITIONAL |
| T3.5 | T3 | T3.4; checked text at T3.5 | CONDITIONAL |
| L2.1 | L2 | contract; checked text at L2.1 | OK |
| L2.2 | L2 | L2.1; checked text at L2.2 | OK |
| L2.3 | L2 | L2.2; checked text at L2.3 | OK |
| L2.4 | L2 | L2.1; checked text at L2.4 | OK |

| Step | Goal | Legal dependencies / justification location | Status |
|---|---|---|---|
| L2.5 | L2 | L2.2, L2.4; checked text at L2.5 | OK |
| L2.6 | L2 | L2.4; checked text at L2.6 | OK |
| L2.7 | L2 | L2.3, L2.6; checked text at L2.7 | OK |
| L2.8 | L2 | L2.4, L2.5, L2.7; checked text at L2.8 | OK |
| L2.9 | L2 | L2.8; checked text at L2.9 | OK |
| P2.1 | P2 | network convention; checked text at P2.1 | CONDITIONAL |
| P2.2 | P2 | L2; checked text at P2.2 | CONDITIONAL |
| P2.2.1 | P2.2 | P2.2, P1; checked text at P2.2.1 | OK |
| P2.2.2 | P2.2 | P2.2, L2; checked text at P2.2.2 | OK |
| P2.2.3 | P2.2 | P2.2.1, P2.2.2; checked text at P2.2.3 | OK |
| P2.2.4 | P2.2 | P2.2.1, P2.2.2, P2.2.3; checked text at P2.2.4 | CONDITIONAL |
| P2.2.5 | P2.2 | P2.2; checked text at P2.2.5 | OK |
| P2.2.6 | P2.2 | P2.2.4, P2.2.5; checked text at P2.2.6 | CONDITIONAL |
| P2.3 | P2 | P2.1, P2.2; checked text at P2.3 | CONDITIONAL |
| T4.1 | T4 | T3; checked text at T4.1 | CONDITIONAL |
| T4.2 | T4 | L1, P2; checked text at T4.2 | CONDITIONAL |
| T4.3 | T4 | T4.1, T4.2, L1; checked text at T4.3 | CONDITIONAL |
| N.1 | N | G; checked text at N.1 | OK |
| N.2 | N | N.1; checked text at N.2 | CONDITIONAL |
| N.O3 | N | source citation; checked text at N.O3 | CONDITIONAL |
| N.3 | N | N.2; checked text at N.3 | CONDITIONAL |
| N.4 | N | S; checked text at N.4 | CONDITIONAL |
| N.O4 | N | source citation; checked text at N.O4 | CONDITIONAL |

# 11 Exact-arithmetic diagnostics

The included `check_exact.py` was rerun for this update. All 173 cases passed using Python's exact `fractions.Fraction` arithmetic: 21 one-dimensional flux cases; 12 product-transport cases ($1 \le n \le 3$, $0 \le k \le 3$); 12 pyramid-flow cases ($2 \le D \le 4$, $1 \le N \le 4$); and 128 quotient recurrence cases ($1 \le D \le 4$, $1 \le N \le 8$, $p \in \{1/5, 1/2, 9/10, 1\}$). The flux check covers $0 \le k \le 20$.

The checks exercise endpoint divergence, coordinate-product weights, conservation, factors of two, lateral/forward energy, and the closed formula against the recurrence. They do not prove universally quantified claims, validate the paper's plotted dataset, or close external citations.

# 12 Minimal clarification plan

Obtain/check EXT-001 and EXT-002 before upgrading the strict true-chain audit verdict. The normalized-sum limit in Remark 3, stray accent command, and unbiased-endpoint wording have been corrected. Retain the finite-mean, empty-product and sufficiently-large-$N$ clarifications when preparing a fully explicit exposition. Check EXT-003/004 separately if their ancillary comparisons require a closed audit. The principle attribution has been corrected to Dirichlet for potentials and Thomson for flows; this correction alone does not close the external-scope check. The supplement is mapped to the authorized corrected paper source.

# Part III

# Reverse dependency audit

## 13 Audit boundary

The primary conclusion is T4: $H_D(N) = \Theta(\alpha^N/N^{D-1})$ for fixed integer $D \geq 1$ and $0 < p < 1$. Separate roots are the quotient formula, quotient order, transport lemma, Remark 3 and ancillary comparisons. The exact source, hypotheses and external boundary are frozen in Part I. Only supplied routes are followed; no alternative finite-state proof is inserted. The leading-constant completion is now part of the corrected source and is audited explicitly.

## 14 Obligation graph

Each row is one primary AND route: every listed dependency is required. Cases $D = 1$ and $D \geq 2$ are exhaustive cases whose proofs are both required, not alternative routes that could hide a failed case. No diagnostic or unprovided rescue route is counted as support.

| Claim | Primary supplied route | AND dependencies |
|---|---|---|
| C0 | T4: combine upper and lower bounds | C1, C2, O0 |
| C1 | T3: hitting-time lower inequality and shorted shells | C3, C4, O1 |
| C2 | T4: hit a face, then apply resistance upper bound | C3, C5, O2 |
| C3 | L1: harmonic capacity and killed Green identities | O3, O4, O5, O6 |
| C4 | P1/T3: shell cuts in series; retain final term | G, O7, O8 |
| C5 | P2: construct unit flow; apply Thomson and sum bound | C6, O6, O9 |
| C6 | P2: one-dimensional path and higher-dimensional pyramid | C7, O10, O11 |
| C7 | L2: coordinate-by-coordinate redistribution | O12, O13, O14 |
| Q0 | T1: unroll differences and telescope | G, O15, O16 |
| Q1 | T2: positive finite series and ratio comparison | Q0, O17, O18 |
| Q2 | R3: proved sharper equivalent | Q1, O19 |
| Q3 | N.2: exact symmetry grouping | O20 |
| Q4 | N.4: predecessor parameter comparison | Q0, O21 |

## 15 Critical findings

O19 is now discharged by the added Remark 3 argument. It is not reachable from the main root C0. O3 and O6 are external-verification leaves on the true-chain route. The local constructive obligations O10–O14 close independently of these citation checks. No cycle, hidden use of the quotient as the true dynamics, or unjustified inference from numerical accuracy was identified.

## 16 Dependency table

| ID | Claim / obligation | Needed by | Route | Classification | Support | Status |
|---|---|---|---|---|---|---|
| C0 | Main true-chain order | root | T4 | root | C1,C2,O0 | open |

| ID | Claim / obligation | Needed by | Route | Classification | Support | Status |
|---|---|---|---|---|---|---|
| C1 | $H_D \geq 2^{-(D-1)}\alpha^N/N^{D-1}$ | C0 | T3 | major claim | C3,C4,O1 | open |
| C2 | $H_D \leq MC'_{D,p}\alpha^N/N^{D-1}$ | C0 | T4 | major claim | C3,C5,O2 | open |
| C3 | Hitting-time identity and two resistance inequalities | C1,C2 | L1 | major claim | O3–O6 | open |
| C4 | Shorted-shell resistance lower bound | C1 | P1,T3 | major claim | G,O7,O8 | discharged locally |
| C5 | Face resistance upper bound | C2 | P2 | major claim | C6,O6,O9 | open |
| C6 | Feasible path/pyramid flow and energy | C5 | P2 | major claim | C7,O10,O11 | discharged locally |
| C7 | Nested-cube transport bound | C6 | L2 | major claim | O12–O14 | discharged |
| G | Shell counts and proposal averages | C4,Q0 | G | proved earlier | G.1–G.6 | discharged |
| Q0 | Exact finite quotient sum | root,Q1,Q4 | T1 | root | G,O15,O16 | discharged |
| Q1 | Quotient asymptotic order | root,Q2 | T2 | root | Q0,O17,O18 | discharged |
| Q2 | Sharper quotient equivalent | root | R3 | root | Q1,O19 | discharged |
| Q3 | Symmetry grouping preserves dynamics | root | N.2 | root | O20 | open |
| Q4 | Predecessor formula correspondence | root | N.4 | root | Q0,O21 | open |
| O0 | Fixed parameters; constants finite and independent of depth | C0 | contracts | side condition | L1.1,P2.2.5 | discharged locally |
| O1 | $\mu(0) = 1$; $(2N-1)^{D-1} \leq (2N)^{D-1}$ | C1 | definitions | algebraic/logical step | T3.4–T3.5 | discharged |
| O2 | $F_N^\circ \subset A_N$ nonempty finite; $\tau_N \leq \tau_{F_N^\circ}$ | C2 | target comparison | definition | V-E; T4.2 | discharged |
| O3 | Obtain/check exact recurrence and fixed-state hitting-time background | C3 | EXT-001 | named theorem | L1.O1 | open |
| O4 | Bounded harmonic potential attains capacity | C3 | energy expansion | proved earlier | L1.2–L1.5; recurrence via O3 | conditional on O3 |
| O5 | Killed Green reversibility, occupation sum, geometric visits | C3 | Green argument | proved earlier | L1.6–L1.8; recurrence via O3 | conditional on O3 |
| O6 | Obtain/check infinite-network duality and unit-flow inequality | C3,C5 | EXT-002 | named theorem | L1.O2 | open |
| O7 | Outer shell separates exterior; finite shell shorting reduces resistance | C4 | shorting route | named theorem | P1,T3.1–T3.2; finite shorting calculation | discharged locally |
| O8 | Cut conductance; series sum and its final term | C4 | series network | algebraic/logical step | P1,T3.3 | discharged |
| O9 | $\sum_{j=1}^{N} \alpha^j/j^{D-1} \leq K\alpha^N/N^{D-1}$ | C5 | eventual ratio | proved earlier | P2.2.5 | discharged |
| O10 | Path/forward/lateral conductances and flow energies | C6 | explicit construction | definition | P2.1,P2.2.1–P2.2.2 | discharged |
| O11 | Current conservation; disjoint energies; dimension and depth cases | C6 | construction closure | side condition | P2.2.3–P2.3 | discharged |
| O12 | Cumulative one-dimensional flux and energy | C7 | cumulative sum | proved earlier | L2.1–L2.3 | discharged |
| O13 | Coordinate-line weights and stage divergence | C7 | product laws | proved earlier | L2.4–L2.7 | discharged |
| O14 | Different coordinate directions; telescoping divergence | C7 | stage sum | algebraic/logical step | L2.8 | discharged |
| O15 | Finite absorption mean; origin boundary equation | Q0 | finite chain | side condition | T1.1; forward clarification | discharged |
| O16 | Recurrence unrolling, product cancellation, final telescoping | Q0 | algebra | algebraic/logical step | T1.2–T1.6 | discharged |
| O17 | $1 \leq F_\ell \uparrow F_\infty < \infty$ | Q1 | positive series | proved earlier | T2.1–T2.2 | discharged |
| O18 | Ratio below one eventually; finite prefix absorbed | Q1 | geometric bound | proved earlier | T2.3–T2.5 | discharged |
| O19 | $Q_D(N)/b_{N-1} \to \alpha/(\alpha-1)$ | Q2 | R3.1–R3.4 | proved earlier | Geometric domination and limits | discharged |
| O20 | Obtain/check exact aggregation theorem and applicability | Q3 | EXT-003 | named theorem | N.O3 | open |
| O21 | Obtain/check prior formula and acceptance convention | Q4 | EXT-004 | unavailable support | N.O4 | open |

# 17 Verdict

**Main true-chain result: INDETERMINATE FROM THE PROVIDED MATERIAL under the frozen external-source boundary**, at O3 and O6. Its local construction is supported, while the invoked external statements remain to be checked. This is not a finding that the theorem is false.

**Exact quotient formula, quotient order, sharper equivalent and transport lemma: FOLLOWS**, with the minor finite-chain and eventual-threshold clarifications in the forward audit.

**Remark 3: FOLLOWS**, with O19 discharged by R3.1–R3.4. The unavailable ancillary external comparisons remain indeterminate at O20/O21.

# A Repository use and provenance

The package contains this PDF and its LaTeX, the corrected manuscript source snapshot, the source hash, the structured step ledger, exact source-label/line mappings, the exact-rational script and results, the unbiased numerical diagnostic, and a small renderer. Run `python3 build.py`, then `pdflatex` twice on `Metropolis_Lamport_Proof_Supplement_Updated.tex`. Run `python3 check_exact.py` for the finite diagnostic suite. Compiling the supplement does not require the paper's figures or bibliography.

The earlier October manuscript source hash began `a41bccc1eb04886d`; it is not the source identity of this update. The source hash above identifies this approved revision; label-to-line mappings are synchronized to its exact bytes. The earlier uploaded PDF predates these authorized edits and is not the source of this supplement.


1. H. Ramachandran and B. Krishnamachari, *Metropolis Random Walks Escape Deep Basins Faster in Higher Dimensions*, corrected LaTeX manuscript, 10 November 2026. Primary proof source.

2. WWresearch, *Lamport Proof*, v0.2.0. https://github.com/WWresearch/lamport-proof/tree/v0.2.0. Conversion, forward audit and reverse dependency audit. The workflows are LLM instructions, not mathematical validation software.

3. L. Lamport, *How to Write a 21st Century Proof*, 2011, revised 2012. https://lamport.azurewebsites.net/pubs/proof.pdf. Methodological reference.

4. P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*, 1984; online version 5 July 2006. Sec. 1.3.5 (finite-network Thomson principle), Sec. 2.2.2 (shorting). These passages were checked in the earlier audit; this update does not claim a newly verified infinite-network convention. https://math.dartmouth.edu/~doyle/docs/walkspdf/walks.pdf.

5. J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, 1976, Sec. 6.3, Thm. 6.3.2. Exact external-text verification remains EXT-003.